

Lens Combinations and Optical Instruments

PHYS 2203 — Introductory Physics III

September 11, 2026

Overview

The last lecture discussed how light behaves at the interface of curved boundaries of media with different refractive indices and established the idea of the thin lens. The lecture discusses how combinations of lenses can be used as *optical instruments*. We develop the governing principle of such combinations (the image formed by one lens acts as the object for the next), work out the special case of two thin lenses in contact, and build the two classic instruments: the compound microscope and the refracting telescope.

1 Recap: The Thin Lens

The previous lecture reduced a thin lens to a single relation between the object distance S , the image distance S' , and the focal length f — the **thin-lens equation**,

$$\boxed{\frac{1}{S} + \frac{1}{S'} = \frac{1}{f}} \quad \text{with transverse magnification} \quad m = -\frac{S'}{S}. \quad (1)$$

The focal length is set by the lens shape through the lensmaker's equation,

$$\frac{1}{f} = \left(\frac{n - n_{\text{air}}}{n_{\text{air}}} \right) \left(\frac{1}{r_1} - \frac{1}{r_2} \right), \quad (2)$$

so a lens with $f > 0$ is **converging** and one with $f < 0$ is **diverging**. The signs follow the usual convention — $S > 0$ for a real object in front of the lens, $S' > 0$ for a real image on the far (transmission) side, $S' < 0$ for a virtual image on the near side; a negative m marks an inverted image and $|m| > 1$ an enlarged one. Every instrument in this lecture is built by applying this one equation, in turn, to each lens of the system.

Image behavior versus object distance

For a converging lens ($f > 0$), solving the thin-lens equation for the image gives

$$S' = \frac{fS}{S - f}, \quad m = -\frac{S'}{S} = \frac{f}{f - S}, \quad (3)$$

or, in the units of Figure 1 (dividing through by f),

$$\frac{S'}{f} = \frac{S/f}{S/f - 1}, \quad m = \frac{1}{1 - S/f}. \quad (4)$$

Sweeping the object distance S (Figure 1) produces two regimes:

- $S > f$ (**the projecting regime**): $S' > 0$ — a *real, inverted* image ($m < 0$), used by cameras and the objective lenses of instruments.
- $0 < S < f$ (**the magnifier regime**): $S' < 0$ — a *virtual, upright, magnified* image ($m > 1$) on the same side as the object. This is how a converging lens is used as a magnifying glass, and it is the case we build on next.

Both S' and m diverge as $S \rightarrow f$ (an object at the focal point images to infinity), and as $S \rightarrow \infty$ the image collapses to $S' \rightarrow f$.

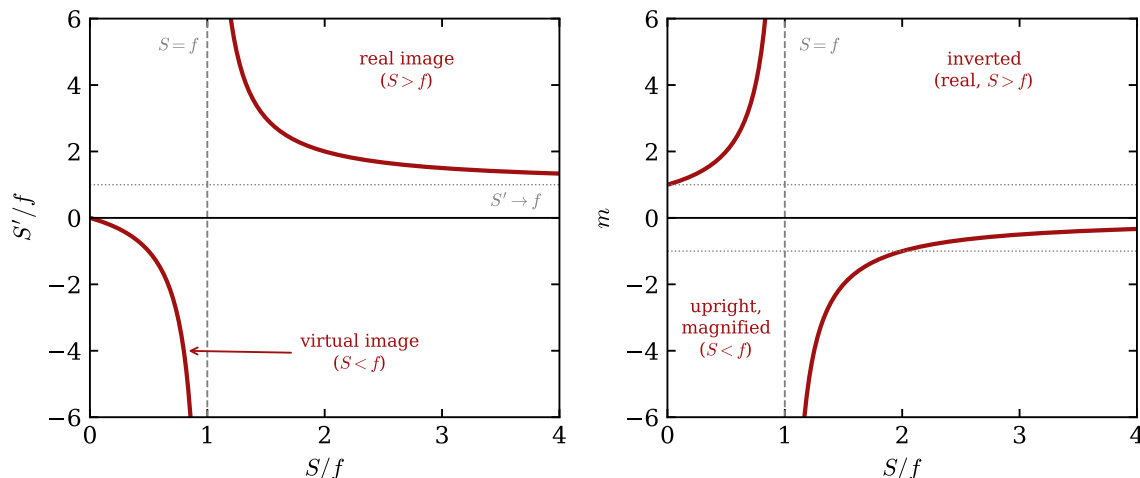


Figure 1: Image distance S' and magnification m as functions of object distance S for a converging lens of focal length $f > 0$, in units of f . Both diverge as $S \rightarrow f$. For $0 < S < f$ the image is virtual, upright, and magnified (the magnifier regime); for $S > f$ it is real and inverted. As $S \rightarrow \infty$, $S' \rightarrow f$.

2 The Eye as Part of the System

The eye is itself a converging lens system: the cornea and the adjustable lens together form a real, inverted image on the retina (the brain handles the inversion). Focusing is done not by moving the lens but by *accommodation*: muscles change the lens's shape, and hence its focal length. Two facts about the eye to know:

- * **Near point** x_{np} : the closest distance between eye and object that still produces a sharp image on the retina. For a normal eye, $x_{np} \approx 25$ cm.
- * Eye strain is lessened when the image approaches infinity (the relaxed eye is focused at infinity).

Correcting the eye: eyeglasses

Eyeglass prescriptions are examples of lens powers (diopters): the power is the reciprocal of the focal length in meters, $P = 1/f$.

- A **nearsighted** (myopic) eye converges too strongly: distant objects focus in front of the retina, and vision is sharp only out to a finite *far point* x_{fp} . The fix is a *diverging* lens that

images an object at infinity to a virtual image at the far point:

$$\frac{1}{\infty} + \frac{1}{-x_{\text{fp}}} = \frac{1}{f} \implies f = -x_{\text{fp}}. \quad (5)$$

For $x_{\text{fp}} = 50 \text{ cm}$: $f = -0.50 \text{ m}$, a prescription of $P = -2.0 \text{ D}$.

- A **farsighted** (hyperopic) eye converges too weakly: its near point lies beyond the normal 25 cm. The fix is a *converging* lens that images an object at 25 cm to a virtual image at the eye's actual near point:

$$\frac{1}{25 \text{ cm}} + \frac{1}{-x_{\text{np}}} = \frac{1}{f}. \quad (6)$$

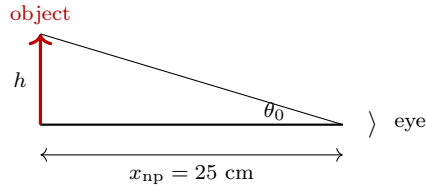
For $x_{\text{np}} = 100 \text{ cm}$: $1/f = 4.0 \text{ D} - 1.0 \text{ D} = 3.0 \text{ D}$, a converging prescription of $P = +3.0 \text{ D}$.

Angular magnification

How big an object *looks* is set by the angle it subtends at the eye, not by its true size — and you enlarge that angle by bringing the object closer. But the eye cannot focus closer than the near point (25 cm), so a magnifier's job is to make an object subtend a large angle while keeping it in focus. We measure the gain with the **angular magnification**, the ratio of the angle subtended with the instrument to the angle subtended without it:

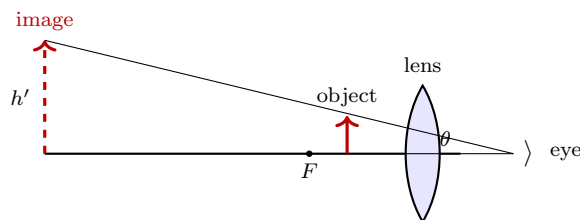
$$M \equiv \frac{\theta}{\theta_0}. \quad (7)$$

Without a magnifier. The largest angle an object can subtend at the eye *and still be seen in focus* is set by the near point: bringing the object closer than x_{np} increases the angle, but the eye can no longer focus. The best the unaided eye can do is therefore the angle subtended at $x_{\text{np}} = 25 \text{ cm}$:



$$\tan \theta_0 = \frac{h}{25 \text{ cm}} \quad \xrightarrow{h \ll 25 \text{ cm}} \quad \theta_0 \approx \frac{h}{25 \text{ cm}}. \quad (8)$$

With a magnifier. Now place the object essentially at the focal point, $S \approx f$. The rays from each object point then leave the lens parallel, sending the image to infinity (from $\frac{1}{S} + \frac{1}{S'} = \frac{1}{f}$ with $S = f$, so $1/S' = 0$). This is the whole trick: an image at infinity lets the *relaxed* eye focus on an object only f away — far inside the 25 cm near point, where the naked eye could never form a sharp image. Through the lens the object subtends the angle it would at distance f :



$$\theta \approx \frac{h}{f}. \quad (9)$$

Result. The lens has supplied *focus*, not extra angle: it lets the eye view the object from f instead of from the 25 cm near-point limit. The angular magnification is the ratio of the two angles,

$$M = \frac{\theta}{\theta_0} = \frac{h/f}{h/(25 \text{ cm})} = \boxed{\frac{25 \text{ cm}}{f}}, \quad (10)$$

so the magnification is larger for a shorter focal length.

3 Combinations of Lenses

Instruments combine two or more lenses. The governing principle:

The image formed by the first lens acts as the object for the second lens — whether that image is real or virtual, and whether or not it actually forms before the second lens intercepts the light.

The thin-lens equation is applied to each lens, with each object and image distance measured from its own lens, and the total lateral magnification is the product $m = m_1 m_2$.

Two thin lenses in contact

For two thin lenses pressed together, the separation is negligible and the image of the first lens is the object of the second with $S_2 = -S'_1$ (remember the sign convention: object and image distances are measured toward opposite sides of the lens, so an image a distance S'_1 behind lens 1 is a *virtual object* at $-S'_1$ for lens 2). Writing the thin-lens equation for each,

$$\frac{1}{S_1} + \frac{1}{S'_1} = \frac{1}{f_1}, \quad \frac{1}{-S'_1} + \frac{1}{S'_2} = \frac{1}{f_2}, \quad (11)$$

and adding, the intermediate distance S'_1 cancels, leaving a single thin-lens equation with an effective focal length

$$\boxed{\frac{1}{f_{\text{eff}}} = \frac{1}{f_1} + \frac{1}{f_2}} \quad \Longleftrightarrow \quad P_{\text{eff}} = P_1 + P_2. \quad (12)$$

Powers of thin lenses in contact simply add — which is why optometrists determine a prescription by stacking trial lenses.

4 The Compound Microscope

A microscope uses two converging lenses in series: an **objective** lens (short focal length f_1) near the specimen, and an **eyepiece** (focal length f_2) near the eye. The two are separated by a tube length ℓ . A single lens would not suffice, because one magnifier is capped at $M = 25 \text{ cm}/f$ and shrinking f enough to give sufficient magnification has limits. Instead, two lenses let their magnifications *multiply* to reach far higher power.

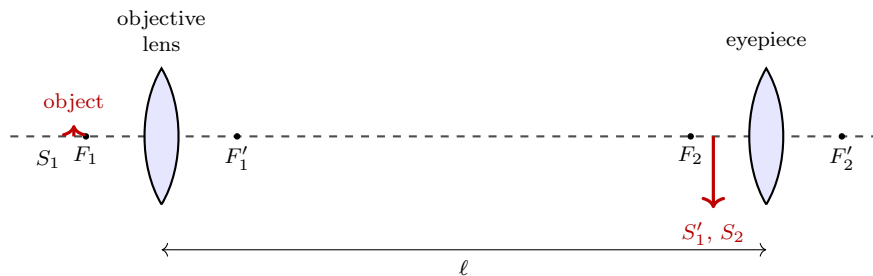


Figure 2: Compound microscope. The objective forms a magnified real image just inside the focal point of the eyepiece; the eyepiece then acts as a simple magnifier on that intermediate image.

For each lens, $\frac{1}{S} + \frac{1}{S'} = \frac{1}{f}$ and $m = -\frac{S'}{S}$.

1) Objective lens — want a large real image. Place the object just outside the focal point F_1 (S_1 slightly greater than f_1). Then

- $S'_1 > 0$ and large ($\approx \ell$) (real image),
- $m_1 \approx -\frac{\ell}{f_1}$ (magnified and inverted; exact when ℓ is measured between the focal points F'_1 and F_2).

2) Eyepiece — a simple magnifier. The intermediate image is placed inside the eyepiece focal length, $0 < S_2 < f_2$. Then

- $S'_2 < 0$ and large (virtual image),
- angular magnification $M = \frac{25 \text{ cm}}{f_2}$.

Total angular magnification. The overall angular magnification is the product of the objective's linear magnification and the eyepiece's angular magnification:

$$\boxed{m_1 M = -\frac{\ell}{f_1 f_2} (25 \text{ cm}).} \quad (13)$$

The magnification is largest when

$$f_1 \text{ small,} \quad f_2 \text{ small,} \quad \ell \text{ large.}$$

5 The Refracting Telescope

A telescope views objects effectively at infinity ($S \approx \infty$), so the object *cannot* be placed near the focal point F_1 as in the microscope. Instead the instrument is built so that the objective's focal point F'_1 coincides with the eyepiece's focal point F_2 . We do this because light from infinity focuses to a real image exactly at F'_1 , and placing that image at the eyepiece's focal point lets the eyepiece — acting as a magnifier — send the rays back out parallel, so a relaxed eye views the distant object comfortably.

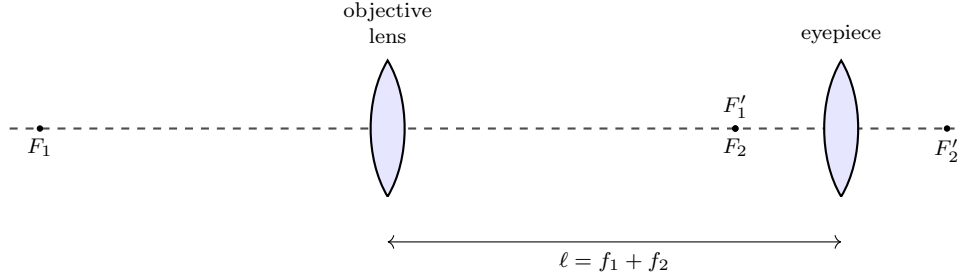


Figure 3: Refracting telescope. The objective forms a real image at its focal point F'_1 , which coincides with the eyepiece's focal point F_2 ; the eyepiece views that image with the relaxed eye.

1) Objective lens — image at F'_1 . With $S_1 \rightarrow \infty$,

$$\frac{1}{S_1} + \frac{1}{S'_1} = \frac{1}{f_1} \implies \frac{1}{\infty} + \frac{1}{S'_1} = \frac{1}{f_1} \implies S'_1 = f_1. \quad (14)$$

2) Eyepiece — a simple magnifier. The image sits at the eyepiece focal point ($S_2 \approx f_2$, with $0 < S_2 < f_2$), sending rays out parallel for the relaxed eye:

$$\frac{1}{f_2} + \frac{1}{S'_2} = \frac{1}{f_2} \implies S'_2 \approx -\infty. \quad (15)$$

Angular magnification. With both the object and the final image at infinity, work with angles. An incoming set of parallel rays at angle θ to the axis is focused by the objective — follow its central ray — to an intermediate image of height $h' = -f_1 \theta$ a distance f_1 behind the objective. The eyepiece, with that image at its focal point F_2 , sends the rays back out parallel at angle $\theta' = h'/f_2$. Hence

$$\boxed{M = \frac{\theta'}{\theta} = -\frac{f_1}{f_2}.} \quad (16)$$

The magnification is largest when

$$f_1 \text{ large} \quad \text{and} \quad f_2 \text{ small.}$$